

GradeOps: Future Goals & Roadmap

1. Advanced AI & ML Workflows

- AI Confidence Highlighting: The AI generates a confidence score for every question. Low-confidence grades are highlighted, guiding the TA to only review answers the AI struggled with.
- Handwriting Translation Overlay: For students with poor handwriting, an overlay shows the AI's transcribed text directly on top of the original PDF.
- RAG-based Past Paper Calibration: Instructors upload previous years' answer keys and graded papers. The AI uses Retrieval-Augmented Generation (RAG) to learn the instructor's specific grading style.
- Batch Processing: Implement FastAPI BackgroundTasks or Celery so users can upload a zip file of PDFs, and the AI grades them in the background without timeout issues.

2. Plagiarism & Integrity

- Plagiarism Network Graph Dashboard: Use a 3D network graph to map out the class. Students are nodes and high similarity scores are links, making cheating rings visually obvious.
- Temporal Analysis: Track upload times versus similarity to catch irregular submission patterns.

3. Instructor & TA Workflow Enhancements

- Interactive Rubric Builder: A drag-and-drop interface for instructors to build dynamic rubrics directly in the UI instead of submitting JSON files.
- PDF Annotations: Allow TAs to drop pins or highlight text on the side-by-side PDF Viewer to show exactly why points were deducted.
- In-App Mentions & Escalation: TAs can tag each other or the Instructor to ask for second opinions on tricky papers.
- Consensus Grading: Configure the system so two TAs must approve a grade before finalizing it. Disagreements escalate to the Instructor.
- TA Leaderboard: A gamified dashboard tracking which TA has reviewed the most papers or has the highest agreement rate with the AI.

4. Student Experience

- Student Dashboard Visualizations: Students can log in to view their results with a Radar Chart comparing their performance against the class average.
- Automated Email Notifications: Integrate SendGrid or Amazon SES to email students a beautifully formatted summary of their score and the AI's justification once a TA approves it.

GradeOps: Future Goals & Roadmap

5. Enterprise & Cloud Linkage

- Cloud Storage Integration: Link with AWS S3, Google Cloud Storage, or Azure Blob Storage to host and serve the uploaded exam PDFs securely and at scale, rather than saving them locally.
- Serverless ML Inference: Deploy the Qwen2-VL model using serverless GPU endpoints (like RunPod or AWS SageMaker) for seamless auto-scaling during peak grading times.
- LMS Integration (Canvas / Moodle): Build an API integration to sync the approved grades directly to the university's Learning Management System.
- Google OAuth / SSO: Allow instructors and students to authenticate securely using their university Google Workspace accounts.

6. Real-time Infrastructure

- WebSockets for Live Dashboard: Implement WebSockets so the Analytics dashboard updates dynamically as the AI grades papers in the background, showing live progress bars and class average shifts.